

RISHABH KARTHIK RAMESH

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Computer Engineering Master's graduate who builds with **large language models** and ships them carefully. Fine-tunes and evaluates **LLMs** in **PyTorch** and **Hugging Face**, builds **retrieval-augmented** and **agentic** systems with **LangGraph** and **FastAPI**, and digs into where models go wrong through **hallucination detection** on financial text. Works across the ML stack in **Python** (Pandas, NumPy, scikit-learn), from cleaning messy data through **CNNs and transformers**, and keeps pace with new GenAI tooling as the field moves.

SKILLS

Languages and Libraries: Python (Pandas, NumPy, scikit-learn), SQL, Bash

ML and Deep Learning: Machine Learning Algorithms, Deep Learning, CNNs, RNNs, Transformers, Model Fine-Tuning, Feature Engineering, Data Preprocessing and Validation, PyTorch, TensorFlow, Keras

Generative AI and LLMs: LLMs (GPT, Claude, Gemini, Llama, local models), Hugging Face (Transformers, Datasets, Tokenizers), Prompt Engineering, AI Workflow Design, RAG, Vector DBs (Chroma), LangChain, LangGraph (agents), LLM Evaluation, GenAI Optimization (cost, latency, performance)

Deployment and Data: FastAPI (async), REST APIs, AWS (Lambda, S3, CloudWatch), Streamlit, PostgreSQL, MySQL, DuckDB, ETL Data Pipelines, Monitoring and Logging, MLOps (familiarity)

Coursework: Docker, OpenShift, Kubernetes

Certifications: Microsoft AI-900, AZ-900, DP-900, SC-900

PROJECTS

Agentic RAG over Financial Filings (LangGraph, FastAPI)

[GitHub ↗](#)

- Built a retrieval-augmented **agent** over SEC 10-K filings using **LangGraph**, **FastAPI**, **Chroma**, and **Hugging Face** embeddings, with a self-checking node that grounds answers in cited passages and abstains when unsupported; on a **27-question** LLM-judged eval, **75%** of answers were fully or partly correct and **86%** of unanswerable questions correctly refused. (Python, LangGraph, FastAPI, Chroma)

LLM Hallucination Detection and Mitigation (AI Safety, Financial Text)

- Built a multi-signal pipeline that catches when an **LLM** makes things up on financial documents, reaching **87%** supported and **79%** hallucinated detection on IBM FinQA, with a **Streamlit** app that flags and corrects unsupported claims for human review. (Python, PyTorch, Hugging Face)

LLM Reasoning Fine-Tuning (Qwen2.5-3B, Local Model)

- Fine-tuned a 3B-parameter open **LLM** with reproducible data-prep and evaluation pipelines (YAML and Bash), benchmarking **49.4%** on GPQA Diamond across reasoning tasks. (Python, PyTorch, Hugging Face)

Serverless Model Inference Service (AWS)

[GitHub ↗](#)

- Put a model behind a **serverless REST endpoint** on **AWS Lambda** that returns predictions in about **270 ms** over **1.3M** records, with monitoring and logging built in. (Python, AWS)

EXPERIENCE

Graduate Teaching Assistant, Virginia Tech

Feb 2025 - May 2025

- Reviewed and **debugged** code for **40+** students on secure data systems, mentoring on data structures, algorithms, and modeling decisions.

Validation Engineer, Victor Tango SAE, Virginia Tech

Sep 2024 - May 2025

- Validated a self-driving **model** across **2** mapped areas, comparing **3** versions on a **PostgreSQL** data system (4,096-point datasets, sub-0.5 s queries) and recommending the one the team shipped.
- Automated test execution and logging and ran iterative **root-cause analysis** on failures and edge cases.

Cybersecurity Analyst L2 (Risk and Data Analytics), Wipro

Apr 2022 - Aug 2024

- Ran **GRC** and **risk analytics** for **2 US-based enterprise clients** in **ServiceNow**, building **Excel** risk-scoring models that helped leadership catch **37%** of critical risks sooner and cut closure time by **25%**.
- Analyzed and reconciled data across multiple systems, surfacing data-quality issues and producing reporting **dashboards** for compliance and security teams.
- Collaborated with cross-functional stakeholders, mentored **2** junior analysts, and maintained process documentation and audit-ready standards.

Data Engineering Intern, TVS Supply Chain Solutions

Jun 2019 - Jul 2019

- Built an **ETL pipeline** pulling live **API** data into **PostgreSQL** with quality checks, plus **Django** dashboards that improved part-recognition accuracy by **24%**.

EDUCATION

M.S., Computer Engineering, Virginia Tech, Blacksburg

May 2026

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Big Data Analytics, Algorithms

B.Tech, Electronics and Communication Engineering, Vellore Institute of Technology

May 2022